

Bridging Distances for Global Participation: Conducting and Theorizing Participatory Design and Research in Hybrid Contexts

Conducting and Theorizing Participatory Design and Research in Hybrid Contexts

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ABSTRACT

Participatory Design (PD) must increasingly be able to serve various global contexts. Overall, a fundamental need has been identified in the PD community to raise previously unheard global voices in design and development work, and to bring various underrepresented stakeholders together. On the other hand, the rapid adoption of various synchronous and asynchronous communication methods through collaborative online tools offers opportunities for more dispersed and more diverse PD research teams, settings and processes. This raises new possibilities for bridging global distances and bringing various stakeholders together for the purposes of conducting multicultural, multi-site PD. Importantly, while there exists extensive research literatures both on PD, on remote ethnography and on Computer-Supported Collaborative Work, we posit that these literatures have not been brought together; nor have the practices and methodologies of these communities been sufficiently jointly explored to support the goals of multisite, multicultural PD. In this workshop, we strive to bridge this gap in knowledge through exploring the opportunities of conducting hybrid multi-site Participatory Design (HMPD) research through synchronous and asynchronous present–telepresent participatory methods, and to share issues, challenges, opportunities, methods and empirical examples pertaining to this as a goal.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; Participatory design.

KEYWORDS

Participatory design, Participatory methodology, Hybrid research settings, Distance work, Remote ethnography, Global South, Global North

ACM Reference Format:

Aale Luusua, Johanna Ylipulli, Dani Kalarikalayil Raju, and Emilia Rönkkö. 2023. Bridging Distances for Global Participation: Conducting and Theorizing Participatory Design and Research in Hybrid Contexts: Conducting and Theorizing Participatory Design and Research in Hybrid Contexts. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3573823>

1 BACKGROUND

In this workshop, we explore the opportunities of conducting multi-site participatory design (PD) and research through hybrid present–telepresent participatory methods, and share issues, challenges, methods and empirical examples pertaining to this as a goal. Participatory Design (PD) must increasingly be able to serve various global contexts. A fundamental need has been identified in the PD community to raise previously unheard global voices in design and development work, and to bring various underrepresented stakeholders together [e.g. 11]. Overall, there is a common creed to work towards a PD that is post-colonial [9], culturally hybrid [8, 10] and which combines various identities [6, 14] and voices from the Global South and North [11].

The overall purpose of PD is to enable the participation of future users and other stakeholders in the design processes of new technologies, in order to both democratize the design process and to co-create new knowledge from various areas of expertise [8, 10, 11]. Owing to the qualitative and context specific nature of PD, the bread and butter of participatory methodology has been physically co-located creative work with participants, with the aim to immerse the practitioners in the use context, and to learn about the practices, needs and values relevant for the stakeholders in this context [e.g. 12]. This embedded, hands-on work is crucial and arguably forms the basis of PD practice. Often, this has also meant that PD research team members have been mostly physically co-located, at least for the duration of the PD process. This approach has fundamental validity, as PD requires an in-depth understanding of the cultural context in which it is performed [8, 10, 11]. Therefore, PD bears many similarities with the ethnographic approach that also has heavily relied on ‘being there’ [13], and these two have been used together and inspired each other [e.g. 12].

Nevertheless, this approach also has its drawbacks, as collocated work means both geographical and financial limitations to the type

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CHI EA '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3573823>

of work that can be accomplished. Conducting multi-site co-located work may be prohibitively time-consuming and expensive, and may also require extensive physical capabilities from both practitioners and participants, leaving many voices unheard. Thus, it could be argued that the traditional co-located methods of PD could be usefully amended with remote methods, and that existing remote methods could be molded to suit the needs of PD.

During the world-wide social distancing efforts of 2020–2022, an enormous leap was made seemingly overnight in adopting novel modalities of distance collaboration, as teams and workplaces became forcibly physically disconnected [1]. This physical separation necessitated distanced virtual team collaboration [1–4]. As we have since moved into a post social distancing world, it is important and timely to seize the lessons learned from this pandemic-induced forced hybridization of collaboration. Furthermore, there is an opportunity to keep developing the potentialities that emerged from the experience. This rapid adoption of various synchronous and asynchronous communication methods through collaborative online tools offers opportunities for more dispersed and more diverse PD research teams, settings and processes. This raises new potentialities for novel spaces for participation [5], bridging global distances and bringing stakeholders of various identities and abilities together for the purposes of conducting multicultural, multi-site PD.

However, this opportunity also raises several questions, including: How can we enable various global stakeholders to take part in global hybrid PD? How can PD practitioners work across cultural spheres in distant settings, and how can we perform culturally aware and immersed work without ‘being there’? What is the role of ‘cultural agents’ in PD processes that span various cultures? What values, competences and awareness do PD practitioners need when conducting PD work in culturally diverse hybrid settings? What can a partly technology-mediated PD process consist of, what knowledge can it produce, and what limitations does it have? What novel challenges and opportunities pertaining to accessibility emerge in a technology-mediated PD process? What PD processes or methods are not suited to hybrid work and best left for the “nonlinear” world? What theories, methods and lessons from the decades of computer-supported work could be applied to PD?

Importantly, while there exists extensive research literatures on PD, on Computer-Supported Collaborative Work and on remote ethnography [13], we recognize a gap at the intersection of these research communities and areas. We posit that these literatures regarding distance work have not been sufficiently brought together with PD research; nor have the practices and methodologies of these communities been sufficiently jointly explored to support the goals of multisite, multicultural PD. In this workshop, we strive to bridge this gap in knowledge and create a novel paradigm of PD through exploring the opportunities of conducting hybrid multi-site Participatory Design (HMPD) research through synchronous and asynchronous present–telepresent participatory methods, and to share issues, challenges, opportunities, methods and empirical examples pertaining to this as a goal.

2 ORGANIZERS

Aale Luusua, D.Sc. (Tech), is an architect, researcher and University Lecturer positioned at the Oulu School of Architecture and

INTERACT Research Unit at the University of Oulu, Finland. With a general research focus on urban environments, citizen participation, digitalization, design theory and interdisciplinary research, Dr. Luusua led the Academy of Finland postdoctoral project *Experiencing Artificial Intelligence in the Smart City (AICity)*. Dr. Luusua’s works have been published in leading international scientific books and journals published by SAGE, Routledge and Springer, and in leading scientific conference proceedings, such as the ACM Conference on Human Factors in Computing Systems and the ACM Conference on Designing Interactive Systems. Dr. Luusua has also co-led international workshops e.g. in the Fifth Decennial Aarhus Conference, Critical Alternatives 2015, at Designing Interactive Systems 2020, and at IndiaHCI2020.

Johanna Ylipulli, Ph.D., is a cultural anthropologist and adjunct professor (docent) in digital culture. She currently works at Aalto University’s Department of Computer Science, where she is the PI of the Academy of Finland Research Fellow project *Digital Inequality in Smart Cities (DISC)* and subproject PI in the project *Designing Inclusive & Trustworthy Digital Public Services for Migrants in Finland*. Her research has focused on cultural and social implications of new digital technologies in urban contexts. Dr. Ylipulli has worked in exceptionally interdisciplinary research environments in leading Finnish Universities, and published broadly, in fora spanning from prominent social scientific journals to flagship conferences of human-computer interaction. These include *International Communication Gazette*, *Technological Forecasting and Social Change* and *Conference on Human Factors in Computing Systems (CHI)*, and publishers such as Routledge and SAGE. Dr. Ylipulli has co-led international workshops e.g. in the Fifth Decennial Aarhus Conference, Critical Alternatives 2015, at Designing Interactive Systems 2020, and at CHI 2021.

Dani Kalarikalayil Raju is co-founder of Studio Hasi, a startup with a mission to diversify future making. He is an alumnus of IDC School of Design, IIT Bombay and has been working with resource constrained user groups (Emergent Users) in India for the past 7 years in collaboration with Computational Foundry at Swansea University. The consistent theme in his work is that of exploring futures with users having constraints such as low literacy, lack of technology experience and low self-efficacy, by engaging them in workshops and deployment studies to re-imagine forward looking technologies for the world. Working with emergent users in two of Mumbai’s slums, his recent work explored the value and uses of photovoltaic (PV) self-powered deformable digital materials for interhome connections. He was awarded Gary Marsden Travel Awards for presenting this work at CHI 2021.

Docent, Dr. **Emilia Rönkkö**, D.Sc. (Tech), is an adjunct professor, architect and docent of urban planning at the Oulu School of Architecture, University of Oulu, Finland. Her expertise covers a wide range of interdisciplinary research topics related to urban health, resiliency strategies and technological innovations supporting smart urban governance. Her primary branch of research work is focusing on evidence-informed urban planning approaches and data-driven tools supporting anticipatory and participatory practices. She currently works as a subproject PI in the Academy of Finland’s Strategic Research project *RECIPE (Resistant Cities: Urban Planning as Means for Pandemic Prevention)*. Amongst its objectives, the project studies virtual collaboration platforms and

their role in knowledge co-creation, knowledge transfer across different boundaries, and overcoming silos in urban planning. In the private sector, Dr. Rönkkö has been involved in the design and development of digital tools for the use of urban planning.

3 WEBSITE

The workshop website is located at: <https://sites.google.com/view/urbanai/conducting-hybrid-pd>

4 PRE-WORKSHOP PLANS

All prospective participants are invited to send the workshop organizers their 2–4 page workshop papers in the ACM Single Column format. The organizers will allow a maximum of 16 papers into the workshop to ensure an engaging, interactive workshop, and a high quality of contributions. For accepted papers, we will seek publication in the INTERACT publications series (cf. 7 Post-Workshop plans).

To further activate the chosen participants prior to the workshop, participants will be asked to submit their introductions on a Miro board in advance. In these introductions, participants are asked to describe their relevant experience and expertise in 5–10 words, and to answer two questions: “*What motivates you to conduct Hybrid & Multi-site Participatory design/research?*” and “*What is the most burning issue related to Hybrid & Multi-site PD for you right now in your research?*” These answers are not circulated outside the workshop.

Participant introductions are shared before the workshop day on a Miro board, and all participants are requested to go through others presentations (slides and/or videos). In this way, participants are familiar with each other already before the workshop day.

5 HYBRID WORKSHOP AND ASYNCHRONOUS ENGAGEMENT

The workshop will be conducted in hybrid format, with participants presenting and working in mixed present–telepresent teams. This approach helps participants to also empathize with the topic at hand: conducting participatory work globally in a hybrid multi-site context. Participants will use online documents and Zoom (or similar) to discuss the group assignments. During the workshop, the groups will build a common Miro board that identifies opportunities and challenges of conducting hybrid, multi-site PD.

This workshop employs asynchronous methods in addition to the typical workshop paper. The aforementioned pre-workshop Miro board will help participants get to know each other beforehand. It will also help participants consider why they want to take part in this workshop, and what are issues of importance to the research community. After the workshop, participants will continue to have access to the board. Analysis and report of the workshop results will be shared on the workshop website in accessible audio-video format as well as in textual form.

6 WORKSHOP STRUCTURE

On the day of the workshop, we begin with an informal round of brief introductions before we settle into the mini-lecture part of the workshop. The hosts will briefly introduce the idea of hybrid and multi-site PD, and a further mini-lecture on the topic of developing

asynchronous and synchronous digital tools for participation in urban planning. These lectures will prepare the participants for the group work. During work session 1, workshop participants will also present their topics of interest. Presentation slides will be collected prior to the workshop to assure a smooth workflow.

For work session 2, we will formulate groups thematically, selected by participants themselves or us, if need be. Groups will receive an assignment to reflect on and discuss relevant theories, case studies, methods, contexts, etc. for the development of the concept of hybrid and multi-site PD. Different groups will get a different assignment, depending on their papers and interests, and all of them will complement each other. The idea is to chart opportunities, challenges, and practical methods to conduct PD in hybrid and in multi-site research settings. The results will be presented on a joint Miro board which will serve as the roadmap to hybrid multi-site Participatory Design.

Finally, for work session 3, groups will present their contributions. To cap the workshop, we will have a closing roundtable discussion to draft a common understanding of the day’s results. Furthermore, we will discuss plans for the future. Our main immediate target is to edit a special issue for INTERACT publication series (cf. Post-workshop plans) and to draft an agenda for developing hybrid and multi-site Participatory Design and research. We will also discuss whether there is interest to edit a Handbook of Hybrid Multi-site Participatory Design.

This is a one-day workshop, with the following preliminary schedule. All times in Central European Time (CET).

Session 1:

10:00am Welcome and hosts’ introductions

10:15am Mini-Lecture on *Experiences, Opportunities and Challenges of Hybrid Multi-site Participatory Design and Research: Case India–Finland*

10:30am Mini-Lecture on *Developing Digital Tools for Participatory Work in Urban Planning*

10:45am Participants’ presentations: Recounting experiences, reflecting on methods and building theory

12:30am Lunch break

Session 2:
01:30pm Identifying opportunities and challenges of Multi-site, Multi-cultural Participatory Design in Hybrid Environments (groups)

02:15pm Identifying tools, methods and theories of Multi-site, Multi-cultural Participatory Design in Hybrid Environments (groups)

03:00pm Coffee break

Session 3:
03:15pm Session 3: Group presentations, Combining the results

04:00pm Session 4: Roundtable discussion; Next steps for the workshop group

05:00pm Closing the workshop

06:00pm Dinner reservation for in-person participants

Light refreshments during breaks are included for in-person participants; however, lunch and dinner are not covered.

7 POST-WORKSHOP PLANS

After the workshop, Workshop participants will be invited to submit a paper to a special issue of the INTERACT publication series (<https://interact.oulu.fi/interact>). Further, opportunities for future

collaborations, co-authoring papers, and editing a Handbook for Hybrid and Multi-site Participatory Design will be planned and discussed at the workshop.

Analysis and report of the workshop results will be later shared on the workshop website in accessible audio-video format as well as in textual form.

8 ACCESSIBILITY

The workshop papers will be published on the conference website as accessible Word documents. Moreover, the participants are required to provide a short, accessible audio-video presentation of their paper. The workshop results are similarly shared on the workshop website as text and in accessible audio-video format.

Fittingly, the workshop itself will be a hybrid event, taking place both in-person and in online environments, which will support accessibility. Some of the hosts will be physically present and some remotely. Due to the hybrid format, participants may use several modalities in the Zoom (or similar) sessions, including verbal discussion, chat function, shared documents, and shared online boards to participate. Importantly, the hybrid nature of the event will also make the workshop financially accessible globally. To this end, we will also only employ free-to-use online tools.

Post-workshop plans also include several accessibility options and modalities. The workshop papers will be published on the workshop website as accessible Word documents. Moreover, the participants are required to provide a short, accessible audio-video presentation of their paper. The workshop results are similarly shared on the workshop website in accessible audio-video format, as well as in textual form.

9 CALL FOR PARTICIPATION

In this workshop, we explore the opportunities of conducting multi-site participatory design (PD) and research through hybrid present-telepresent participatory methods, and share issues, challenges, methods and empirical examples pertaining to this as a goal.

PD must increasingly be able to address and serve various global contexts. Overall, then, there is a fundamental need to raise previously unheard global voices in technology design and development, and to bring various stakeholders together. The bread and butter of participatory methodology has been physically co-located creative work with participants, with the aim to immerse the practitioners in the use context, and to learn about the practices, needs and values relevant for the stakeholders in this context. This embedded, hands-on work is crucial and arguably forms the basis of PD practice. Often, this has also meant that PD research teams have been mostly physically co-located, at least for the duration of the PD process. This approach has fundamental validity, as PD requires an in-depth understanding of the cultural context in which it is performed.

Nevertheless, this approach also has its drawbacks, as collocated work means both geographical and financial limitations to the type of work that can be accomplished. Conducting multi-site co-located work may be prohibitively time-consuming and expensive, and may also require extensive physical capabilities from both practitioners and participants, leaving many voices unheard. Furthermore, the

rapid adoption of various synchronous and asynchronous communication methods through online collaborative tools offer opportunities for more continuous, more dispersed, and more diverse PD research settings and processes.

During the world-wide social distancing efforts of 2020–2022, an enormous leap forward was made seemingly overnight in adopting novel modalities of participation and collaboration. As we have since moved into a post social distancing world, it is important and timely to both seize the lessons learned from the pandemic-induced forced hybridization of PD, and to keep developing the potentialities that emerged from the experience. However, this opportunity also raises several questions, including: How can we enable various global stakeholders to take part in global hybrid PD? How can PD practitioners work across cultural spheres in distant settings, and how can we perform culturally aware and sensitive work without 'being there'? What can a partly technology-mediated PD process consist of, what knowledge can it produce, and what limitations does it have? What novel challenges and opportunities pertaining to accessibility emerge in a technology-mediated PD process? What values, competences and awareness do PD practitioners need when conducting PD work in culturally diverse hybrid settings? What PD processes or methods are not suited to hybrid work and best left for the "nonline" world? What theories, methods and lessons from other fields of research, e.g. from remote ethnography or Computer-Supported Collaborative Work, could be applied?

We invite those interested to submit a position paper (3–4 pages) in the SIGCHI Single column paper format. These contributions may address, for example:

- Theoretical considerations: e.g., utilising key theoretical concepts from various fields to build understanding of how to conduct hybrid participatory design and research in various global contexts
- Case studies: empirical works that analyze experiences of hybrid participatory design and research in various global contexts
- Design-based explorations in real-world settings: e.g. implementation of participatory design and research approaches in different global contexts in hybrid environments
- Methods: Methodologies that can contribute towards better implementation of participation in various global contexts and in hybrid environments
- Thematic issues; e.g. ethical issues; various user or stakeholder groups, various areas of design, global perspectives, etc. pertaining to conducting hybrid PD in multi-side, multi-cultural contexts

Deadline: Feb 8, 2023 (23:59 AoE)

Notifications: Mar 1, 2023 (23:59 AoE)

Workshop Duration: One day

Papers should be sent to: aale.luusua(at)oulu.fi

ACKNOWLEDGMENTS

The authors would like to acknowledge the valuable input of co-design and research participants in Finland and India. This work was supported by the Academy of Finland funded project Digital Inequality in Smart Cities, DISC, (332143).

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